

STANDARD 2

Stage 6

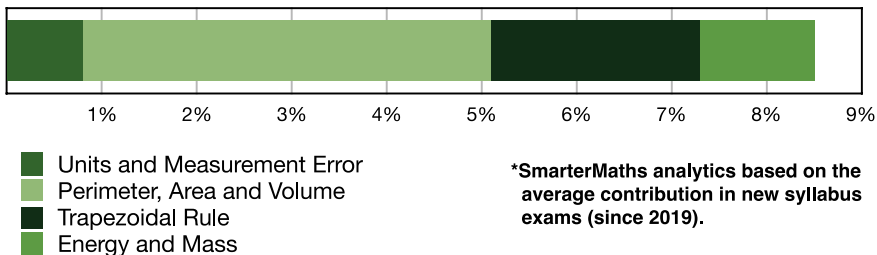
Measurement (Std 2), M1 Applications of Measurement (Y11)

Perimeter, Area and Volume (Std 2)

Teacher: M K

Exam Equivalent Time: 76.5 minutes (based on allocation of 1.5 minutes per mark)

M1 Applications of Measurement



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- M1 Applications of Measurement* has contributed a healthy average of 8.5% per new syllabus Std2 exam since it was introduced in 2019.
- We have split this area into 4 categories for the purpose of analysis: 1-Units and Measurement Error (0.8%), 2-Perimeter, Area and Volume (4.3%), 3-Trapezoidal Rule (2.2%) and 4-Energy and Mass (1.2%).
- This analysis looks at *Perimeter, Area and Volume*.

ANALYSIS - Common pitfalls

- Perimeter, Area and Volume* has contributed 5 separate longer answer questions in the 2022-24 period, regularly producing sub-50% mean marks. This is after the 2021-2019 Std2 papers presented low difficulty examinations of the topic area.
- This topic area is very broad and to revise effectively, there is no substitute to doing lots of varied past exam questions.
- Well designed revision must include a focus on area and volume questions that incorporate circular measure which are easily the most commonly asked and poorly answered Most recently 2024 Std2 34 illustrates this key revision example.
- Water catchment questions (examined in 2022 for the first time since 2016) have caused problems and deserve attention.

Questions

1. Measurement, STD2 M1 2010 HSC 28b

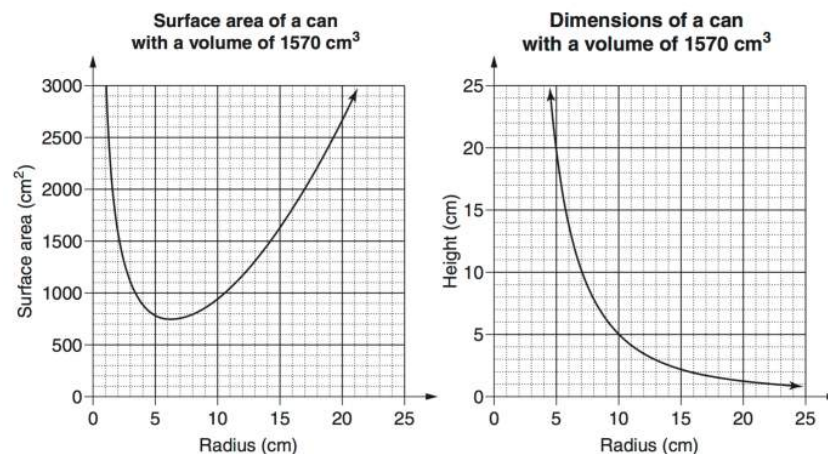
- ⚠ Part i: RAP Data (vs state mean) - Bottom 3%: School result (21%) was 6% below state average (27%)**
⚠ Part i: RAP Data (raw) - School result was 21%

Moivre's manufacturing company produces cans of Magic Beans. The can has a diameter of 10 cm and a height of 10 cm.



- Cans are packed in boxes that are rectangular prisms with dimensions 30 cm × 40 cm × 60 cm. What is the maximum number of cans that can be packed into one of these boxes? **(1 mark)**
- The shaded label on the can shown wraps all the way around the can with no overlap. What area of paper is needed to make the labels for all the cans in this box when the box is full? **(2 marks)**
- The company is considering producing larger cans. Monica says if you double the diameter of the can this will double the volume. Is Monica correct? Justify your answer with suitable calculations. **(2 marks)**

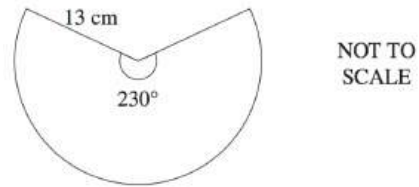
The company wants to produce a can with a volume of 1570 cm³, using the least amount of metal. Monica is given the job of determining the dimensions of the can to be produced. She considers the following graphs.



- iv. What radius and height should Monica recommend that the company use to minimise the amount of metal required to produce these cans? Justify your choice of dimensions with reference to the graphs and/or suitable calculations. **(2 marks)**

2. Measurement, STD2 M1 2012 HSC 27b

The sector shown has a radius of 13 cm and an angle of 230° .

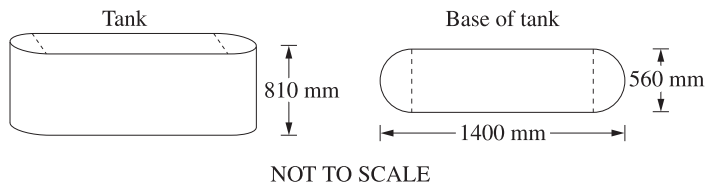


What is the perimeter of the sector to the nearest centimetre? **(2 marks)**

3. Measurement, STD2 M1 2014 HSC 27c

The base of a water tank is in the shape of a rectangle with a semicircle at each end, as shown.

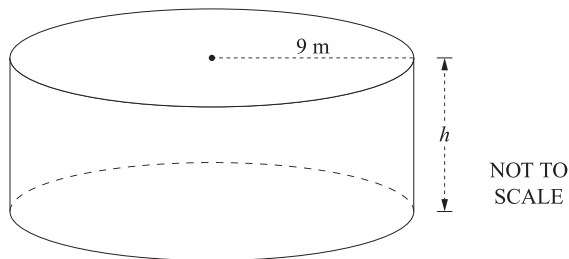
The tank is 1400 mm long, 560 mm wide, and has a height of 810 mm.



What is the capacity of the tank, to the nearest litre? **(4 marks)**

4. Measurement, STD2 M1 2018 HSC 30a

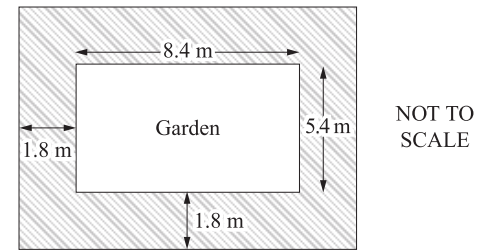
A cylindrical water tank has a radius of 9 metres and a capacity of 1.26 megalitres.



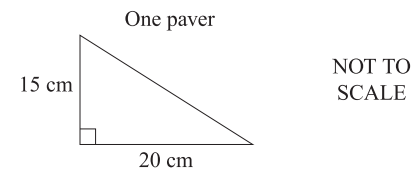
What is the height of the water tank? Give your answer in metres, correct to two decimal places. **(3 marks)**

5. Measurement, STD1 M1 2019 HSC 36

A path 1.8 m wide is being built around a rectangular garden. The garden is 8.4 m long and 5.4 m wide. The path is shaded in the diagram.



The path is to be covered with triangular pavers with side lengths of 15 cm and 20 cm as shown.



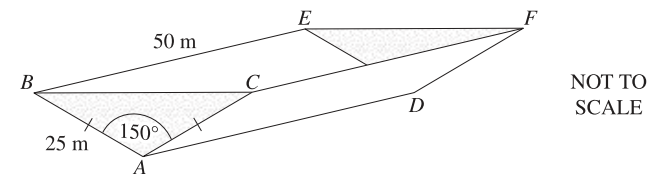
The pavers are to be laid to cover the path with no gaps or overlaps.

How many pavers are needed? **(4 marks)**

6. Measurement, STD2 M1 2022 HSC 28

A dam is in the shape of a triangular prism which is 50 m long, as shown.

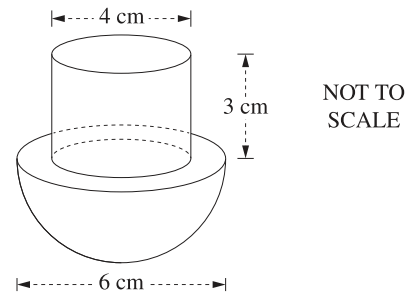
Both ends of the dam, ABC and DEF , are isosceles triangles with equal sides of length 25 metres. The included angles BAC and EDF are each 150° .



Calculate the number of litres of water the dam will hold when full. **(4 marks)**

7. Measurement, STD2 M1 2022 HSC 34

A composite solid is shown. The top section is a cylinder with a height of 3 cm and a diameter of 4 cm. The bottom section is a hemisphere with a diameter of 6 cm. The cylinder is centred on the flat surface of the hemisphere.



Find the total surface area of the composite solid in cm², correct to 1 decimal place. (4 marks)

8. Measurement, STD2 M1 2024 HSC 34

A container for soccer balls is made using two half spheres joined to each end of a cylindrical body.



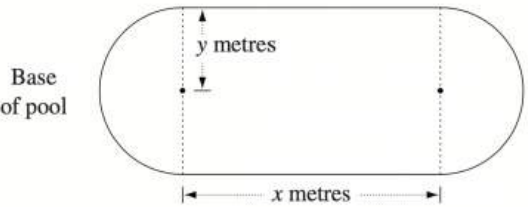
Three soccer balls fit exactly inside the container. Each ball has a diameter of 23 cm.

The hemispherical ends of the container just touch the surface of the soccer balls.

What is the total surface area of the container? Give your answer in square metres correct to one decimal place. (4 marks)

9. Measurement, STD2 M1 2005 HSC 28a

The Mitchell family has moved to a new house which has an empty swimming pool. The base of the pool is in the shape of a rectangle, with a semicircle on each end.



i. Explain why the expression for the area of the base of the pool is $2xy + \pi y^2$. (1 mark)



The pool is 1.1 metres deep.

ii. The sides and base of the pool are covered in tiles. If $x = 6$ and $y = 2.5$, find the total area covered by tiles. (Give your answer correct to the nearest square metre.) (4 marks)

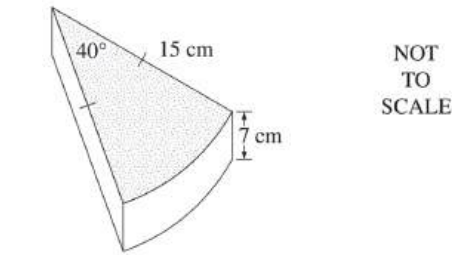
Before filling the pool, the Mitchells need to install a new shower head, which saves 6 litres of water per minute.

The shower is used 5 times every day, for 3 minutes each time.

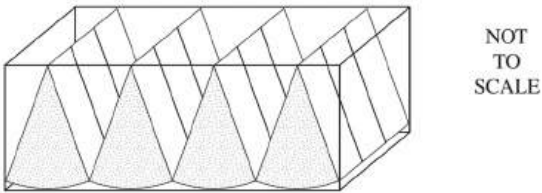
iii. If the charge for water is \$1.013 per kilolitre, how much money would be saved in one year by using this shower head? (Assume there are 365 days in a year.) (2 marks)

10. Measurement, STD2 M6 2008 HSC 25c

Pieces of cheese are cut from cylindrical blocks with dimensions as shown.

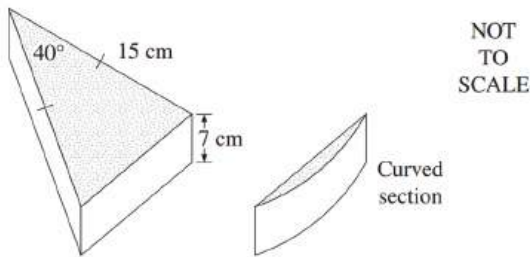


Twelve pieces are packed in a rectangular box. There are three rows with four pieces of cheese in each row. The curved surface is face down with the pieces touching as shown.



i. What are the dimensions of the rectangular box? (4 marks)

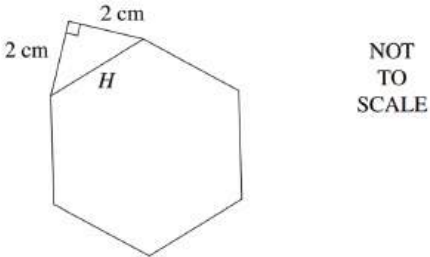
To save packing space, the curved section is removed.



ii. What is the volume of the remaining triangular prism of cheese? Answer to the nearest cubic centimetre. (2 marks)

11. Algebra, STD2 A1 2007 HSC 28b

This shape is made up of a right-angled triangle and a regular hexagon.



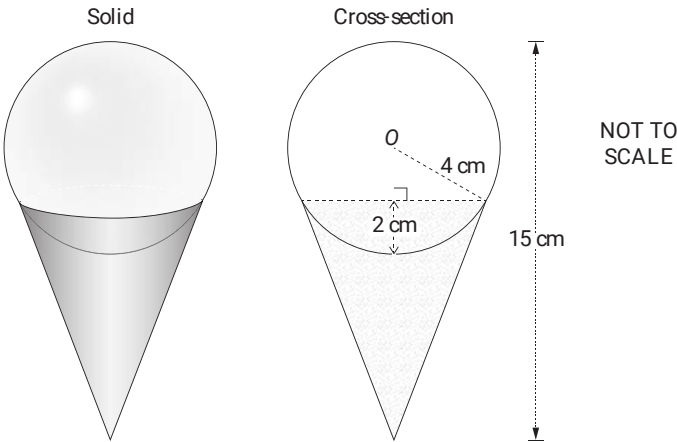
The area of a regular hexagon can be estimated using the formula $A = 2.598H^2$ where H is the side-length.

Calculate the total area of the shape using this formula. (3 marks)

12. Measurement, STD2 M1 2017 HSC 30e

A solid is made up of a sphere sitting partially inside a cone.

The sphere, centre O , has a radius of 4 cm and sits 2 cm inside the cone. The solid has a total height of 15 cm. The solid and its cross-section are shown.



Using the formula $V = \frac{1}{3}\pi r^2 h$ where r is the radius of the cone's circular base and h is the perpendicular height of the cone, find the volume of the cone, correct to the nearest cm^3 ? (3 marks)

Worked Solutions

1. Measurement, STD2 M1 2010 HSC 28b

i. Maximum # Cans = $4 \times 3 \times 6$
 $= 72$ cans

♦♦ Mean mark 27%

ii. Label Area (1 can) = $2\pi rh$
 $= 2 \times \pi \times 5 \times 9$
 $= 90\pi$
 $= 282.7433 \dots \text{ cm}^2$

♦ Mean mark 38%
MARKER'S COMMENT: Many students didn't account for the clearance of 0.5 cm at the top and bottom of each can.

$\therefore$ Label Area (72 cans)
 $= 72 \times 282.7433 \dots$
 $= 20\,357.52 \dots$
 $= 20\,358 \text{ cm}^2$ (nearest cm^2)

iii. Original volume = $\pi r^2 h$
 $= \pi \times 5^2 \times 10$
 $= 785.398 \dots \text{ cm}^3$

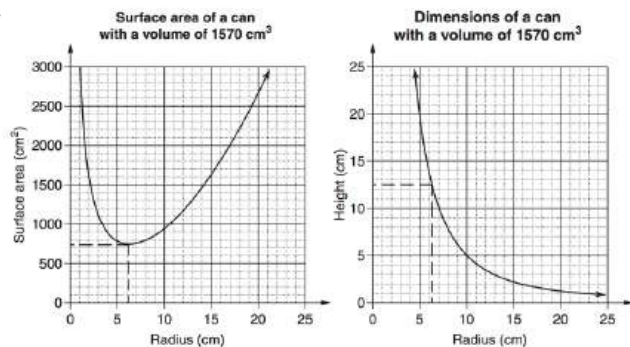
If the diameter doubles, radius = 10 cm

New volume = $\pi \times 10^2 \times 10$
 $= 3141.592 \dots \text{ cm}^3$

♦ Mean mark 44%
MARKER'S COMMENT: Many students performed calculations in this part without concluding if Monica is correct or not. Read the question carefully.

$\therefore$ Monica is incorrect because the volume doesn't double. It increases by a factor of 4.

iv.



♦♦ Mean mark 26%

Minimum metal used when surface area is a minimum.
 From graph, minimum surface area when $r = 6.3$ cm

Worked Solutions

When $r = 6.3$ cm, $h = 12.6$ cm (from graph)

$\therefore$ She should recommend radius 6.3 cm and height 12.6 cm

2. Measurement, STD2 M1 2012 HSC 27b

Perimeter = $2 \times \text{radius} + \text{arc length}$

Arc length = $\frac{\theta}{360} \times 2 \times \pi \times r$
 $= \frac{230}{360} \times 2 \times \pi \times 13$
 $= 52.1853 \dots$

♦ Mean mark 39%
MARKER'S COMMENT: The formula of the length of an arc is given in the formula sheet.

$\therefore$ Perimeter = $2 \times 13 + 52.1853 \dots$
 $= 78.1853 \dots$
 $= 78$ cm (nearest cm)

3. Measurement, STD2 M1 2014 HSC 27c

$V = Ah$

Finding Area of base

Semi-circles have radius 280 mm = 0.28 m

$\therefore$ Area of 2 semicircles

$= 2 \times \frac{1}{2} \times \pi r^2$
 $= \pi \times (0.28)^2$
 $= 0.2463 \dots \text{ m}^2$

♦ Mean mark 41%
STRATEGY: Adjusting measurements to metres makes the final conversion to litres simple.

Area of rectangle

$= l \times b$
 $= (1.4 - 2 \times 0.28) \times 0.56$
 $= 0.4704 \text{ m}^2$

$\therefore$ Volume = Ah
 $= (0.2463 \dots + 0.4704) \times 0.810$
 $= 0.580527 \dots \text{ m}^3$
 $= 580.527 \dots \text{ L}$ (using $1 \text{ m}^3 = 1000 \text{ L}$)
 $= 581 \text{ L}$ (nearest L)

4. Measurement, STD2 M1 2018 HSC 30a

Converting megalitres to m³ (using 1 m³ = 1000 L):

$$\begin{aligned} 1.26 \text{ ML} &= \frac{1.26 \times 10^6}{10^3} \\ &= 1.26 \times 10^3 \text{ m}^3 \\ &= 1260 \text{ m}^3 \end{aligned}$$

♦ Mean mark 48%.

$$\begin{aligned} V &= \pi r^2 h \\ 1260 &= \pi \times 9^2 \times h \\ h &= \frac{1260}{\pi \times 9^2} \\ &= 4.951 \dots \\ &= 4.95 \text{ m (2 d.p.)} \end{aligned}$$

5. Measurement, STD1 M1 2019 HSC 36

$$\begin{aligned} \text{Shaded Area} &= \text{Large rectangle} - \text{garden area} \\ &= (1.8 + 8.4 + 1.8) \times (1.8 + 5.4 + 1.8) - (8.4 \times 5.4) \\ &= 12 \times 9 - 8.4 \times 5.4 \\ &= 62.64 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of 1 paver (in m}^2\text{)} &= \frac{1}{2} \times 0.15 \times 0.20 \\ &= 0.015 \text{ m}^2 \end{aligned}$$

♦♦ Mean mark 18%.
COMMENT: Convert all measurements to the same units to minimise errors (either m² or cm²).

$$\begin{aligned} \therefore \text{Number of pavers needed} &= \frac{62.64}{0.015} \\ &= 4176 \end{aligned}$$

6. Measurement, STD2 M1 2022 HSC 28

$$\begin{aligned} V &= Ah \\ \text{Use sine rule to find } A: \end{aligned}$$

♦♦ Mean mark 33%.

$$\begin{aligned} A &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 25 \times 25 \times \sin 150^\circ \\ &= 156.25 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \therefore V &= 156.25 \times 50 \\ &= 7812.5 \text{ m}^3 \end{aligned}$$

Since 1 m³ = 1000 litres:

$$\begin{aligned} \text{Dam capacity} &= 7812.5 \times 1000 \\ &= 7\,812\,500 \text{ L} \end{aligned}$$

7. Measurement, STD2 M1 2022 HSC 34

$$\begin{aligned} \text{S.A. of Cylinder} &= \pi r^2 + 2\pi rh \\ &= \pi(2^2) + 2\pi(2)(3) \\ &= 16\pi \text{ cm}^2 \end{aligned}$$

♦ Mean mark 50%.

$$\begin{aligned} \text{S.A. of Hemisphere} &= \frac{1}{2} \times 4\pi r^2 \\ &= 2\pi(3^2) \\ &= 18\pi \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of Annulus} &= \pi R^2 - \pi r^2 \\ &= \pi(3^2) - \pi(2^2) \\ &= 5\pi \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Total S.A.} &= 16\pi + 18\pi + 5\pi \\ &= 39\pi \\ &= 122.522 \dots \\ &= 122.5 \text{ cm}^2 \text{ (to 1 d.p.)} \end{aligned}$$

8. Measurement, STD2 M1 2024 HSC 34

Ball diameter = 23 cm $\Rightarrow r = 11.5$ cm

S.A. (2 half spheres) = $2 \times \frac{1}{2} \times 4\pi \times 11.5^2$
= 1662 cm²

Cylinder width = ball circumference = $2 \times \pi \times 11.5$

Cylinder height = $2 \times 23 = 46$ cm

S.A. (cylinder) = $2\pi rh = 2 \times \pi \times 11.5 \times 46 = 3324$ cm²

Note: 1 m² = 100 cm $\times$ 100 cm = 10 000 cm²

S.A. (container) = 1662 + 3324
= 4986 cm²
= $\frac{4986}{10\,000}$ m²
= 0.5 m² (1 d.p.)

◆ Mean mark 41%.

9. Measurement, STD2 M1 2005 HSC 28a

i. Area of base = Area of rectangle +
 $2 \times$ Area of semi-circle
= $(x \times 2y) + 2 \times \left(\frac{1}{2} \times \pi \times y^2\right)$
= $2xy + \pi y^2$

ii. Area of base = $(2 \times 6 \times 2.5) + (\pi \times 2.5^2)$
= 49.634... m²

Area of walls = Length $\times$ Height

Length = $2x + 2 \times$ semi-circle perimeter
= $(2 \times 6) + 2 \times \left(\frac{1}{2} \times 2 \times \pi \times 2.5\right)$
= 12 + 15.707...
= 27.707... m

$\therefore$ Area of walls = 27.707×1.1
= 30.478... m²

$\therefore$ Total Area covered by tiles
= 49.634... + 30.478...
= 80.11...
= 80 m² (nearest m²)

iii. Water saved = $5 \times 3 \times 6$
= 90 L per day.

Water saved per year
= 90×365
= 32 850 L
= 32.85 kL

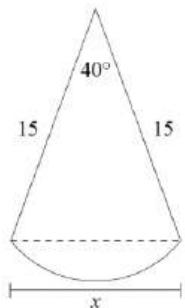
$\therefore$ Money saved = $32.85 \times \$1.013$
= \$33.277...
= \$33.28 (nearest cent)

10. Measurement, STD2 M6 2008 HSC 25c

- i. Box height = 15 cm
(radius of the arc)

$$\begin{aligned}\text{Box width} &= 3 \times 7 \\ &= 21 \text{ cm}\end{aligned}$$

$$\text{Box length} = 4x$$



Using cosine rule

$$\begin{aligned}c^2 &= a^2 + b^2 - 2ab\cos C \\ x^2 &= 15^2 + 15^2 - 2 \times 15 \times 15 \times \cos 40^\circ \\ &= 450 - 344.7199\dots \\ &= 105.2800\dots \\ x &= 10.2606\dots\end{aligned}$$

$$\begin{aligned}\text{Box length} &= 4 \times 10.2606\dots \\ &= 41.04\dots\end{aligned}$$

$$\therefore \text{Dimensions are } 41 \text{ cm} \times 21 \text{ cm} \times 15 \text{ cm}$$

- ii. Volume = Ah

$$h = 7 \text{ cm}$$

Find A :

$$\begin{aligned}A &= \frac{1}{2}ab\sin C \\ &= \frac{1}{2} \times 15 \times 15 \times \sin 40^\circ \\ &= 72.3136\dots\end{aligned}$$

$$\begin{aligned}\therefore V &= 72.3136\dots \times 7 \\ &= 506.195\dots \\ &= 506 \text{ cm}^3 \text{ (nearest whole)}\end{aligned}$$

◆ Mean mark 45%.

◆◆◆ Mean mark 22%.

11. Algebra, STD2 A1 2007 HSC 28b

$$\text{Area} = 2.598H^2$$

Using Pythagoras

$$\begin{aligned}H^2 &= 2^2 + 2^2 \\ &= 8\end{aligned}$$

$$H = \sqrt{8}$$

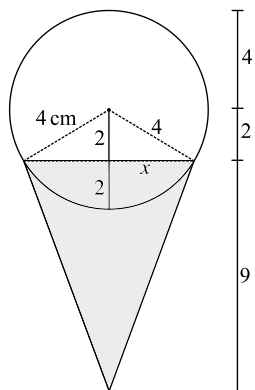
$$\begin{aligned}\therefore \text{Area of hexagon} &= 2.598 \times (\sqrt{8})^2 \\ &= 20.784 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of triangle} &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 2 \times 2 \\ &= 2 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\therefore \text{Total Area} &= 20.784 + 2 \\ &= 22.784 \text{ cm}^2\end{aligned}$$

12. Measurement, STD2 M1 2017 HSC 30e

$$V = \frac{1}{3} \times \text{base of cone} \times \text{height}$$



Consider the circular base area of the cone,

Find x (using Pythagoras):

$$x^2 = 4^2 - 2^2 = 16 - 4 = 12$$

$$x = \sqrt{12} \text{ cm}$$

$$\therefore V = \frac{1}{3} \times \pi \times \left(\sqrt{12}\right)^2 \times (15 - 6)$$

$$= \frac{1}{3} \times \pi \times 12 \times 9$$

$$= 113.097\dots$$

$$= 113 \text{ cm}^3 \text{ (nearest cm}^3\text{)}$$